



R.K.D.F. UNIVERSITY, BHOPAL

M.TECH (Construction Technology & Management)

First YEAR (Semester – 2)

Course Content & Grade

Branch	Subject Title	Subject Code	Contact Hours per Week	Total Credits
M.TECH (Construction Technology & Management)	Contract Management-	MTCTM-2001	3L-1T-0P	4

Unit-I Contract Management-I

Types of Construction contract, Lumpsum, Unit rate, cost plus-fee, Cost Plus percentage fee, Incentive Contracts, Nature of Contract, Contract Documents and Contracting procedures, contract revisions.

Unit-2 Contract Management-II

Technical Specifications, Drawings, Tender Bond, Labour and Material Payment Bonds, Scrutiny of Tenders, acceptance, letter of indent. Important Contract clauses, Terms of Payment, retention acceptance and final payment, maintenance period, Time for Completion, Extension of time, Variation in work and conditions, claims and disputes, liquidated damages, Termination rights and responsibility of client, Architect, Engineer, Contractor, Professional liability.

Unit-3 Tender Management

Advanced Techniques of Estimating, Principles of Analysis of rates and Specification, writing for different types of construction industries, capital structure, Theories.

Unit-4 Legal Framework of Construction

Constitutional provisions relating to Business and industry, Master Plans, Indian Contract Act. Arbitration Act.

Unit-5 Labour Laws and Legislation

Contract Labour (RRA) Act 1970, laws relating to wages, bonus & industrial disputes.

References:-

1. Construction Management by Daniel W. Halpin & Bolivar A. Senior
2. Construction Management and Planning by B. Sengupta & H. Guha
3. Modern Construction Management by Frank Harris and Ronald McCaffer
4. Principles of Construction Management by Roy Pilcher
5. Construction Management by Paul Watson



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Branch	Subject Title	Subject Code	Contact Hours per Week	Total Credits
M.TECH (Construction Technology & Management)	Pre Engineering Construction and Technology	MTCTM-2002	3L-1T-0P	4

Course objectives:

This course will enable students to

CO1:-Understand various principles of prefabrication

CO2:-Understand the type of prefabricated elements

CO3:-Understand the method of hoisting technology in prefabrication.

CO4:-Understand the method of Precast sandwich Panels in prefabrication

CO5:-Understand the basic construction of the pre engineered buildings

Unit-1 General Principles of PreFabrication Comparison with monolithic construction –

Types of prefabrication – site and plant prefabrication - Economy of prefabrication – Modular coordination – Standardization – Planning for Components of prefabricated structures –

Disuniting of structures – Design of simple rectangular beams and I beams – Handling and erection stresses – Elimination of erection stresses – Beams, columns – Symmetrical frames.

Unit-2 Prefabricated Elements Roof and floor panels, ribbed floor panels – wall panels – footings – Joints for different structural Connections – Effective sealing of joints for water proofing – Provisions for non-structural fastenings –Expansion joints in pre-cast construction.

Designing and detailing of precast unit for factory structures –Purlins, Principal rafters, roof trusses, lattice girders, gable frames – Single span single storeyed frames –Single storeyed buildings – slabs, beams and columns.

Unit-3 Production and Hoisting Technology Choice of production setup – Manufacturing methods – Stationary and mobile production – Planning of production setup – Storage of precast elements – Dimensional tolerances – Acceleration of concrete hardening. Equipments for hoisting and erection – Techniques for erection of different types of members like Beams, Slabs, Wall panels and Columns – Vacuum lifting pads .

Unit-4 Precast sandwich Panels ,Prestressed concrete solid flat slabs, Hollow core slab/panels, Prestressed concrete Double “T”, Bridge, Precast segmental Box Girders, Specifications and Seismic considerations



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Unit-5

Pre-Engineered Buildings Introduction – Advantages - Pre Engineered Buildings Vs Conventional Steel Buildings - Design of Pre Engineered Buildings (PEB) – Applications

Course outcomes:

On completion of this course, students are able to:

- To design the pre-engineered structures and execute the same for a given structure
- To know the different types of stresses acting on the structures while lifting the prefabricated structures and type of equipment required to support such stresses

REFERENCES:

1. L. Mokka, “Prefabricated Concrete for Industrial and Public Structures,” Publishing House of the Hungarian Academy of Sciences, Budapest, 2007.
2. T. Koncz, “Manual of Precast Concrete Construction”, Vol. I, II, III & IV, Berlin, 1971.
3. B. Lewicki, “Building with Large Prefabricates”, Elsevier Publishing Company, Amsterdam, London, New York, 1998.
4. Structural Design Manual, Precast Concrete Connection Details, Society for the
5. Studies in the use of Precast Concrete, Netherland Betor Verlag, 2009.
6. Hass, A.M. Precast concrete design and Applications, Applied Science Publishers, 1983.



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First YEAR (Semester – 2)

Course Content & Grade

Branch	Subject Title	Subject Code	Contact Hours per Week	Total Credits
M.TECH (Construction Technology & Management)	Construction Equipment and Material Management	MTCTM-2003	3L-1T-0P	4

Course Objective:-

This course will enable students to

CO1:-Understand various Planning and Selection of Construction Equipment

CO2:-Understand the production estimates of various construction equipment.

CO3:-Understand the Economics of Construction Equipment.

CO4:-Understand the System Approach Problems of equipment management.

CO5:-Understand the concept of material management.

Unit-1 Planning and Selection of Construction Equipment

Advantage of mechanization of Construction industry. Merits of Labour intensive construction. Planning for construction equipment's. Analytical studies, equipment operation. Selection of construction machinery & equipment.

Unit-2 Production Estimates, Sizing and Matching Cycle time capacity ratings and output of Excavators, Power shovels, drag lines, scrapper, bulldozers, tractor shovels rippers, motor graders etc. Sizing and matching. Capacity ratings and output of compactors, aggregate processing plant concrete production plants etc.

Unit-3 Economics of Construction Equipment

Equipment working rates, Investment cost, Depreciation cost, major repair cost. Cost of fuel and lubricants .Cost of labour, servicing and field repairs, overheads. Recommendations of statutory bodies.

Unit-4 System Approach Problems of equipment management. Application of CPM in equipment management. Application of the assignment model, transportation model and waiting line models in equipment management.

Unit-5 Material Management

Materials planning and budgeting. Role and functions at different levels of management and budgeting variations. Stages of materials management. A.B.C. analysis. Advantages,



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mechanics purpose cautions, limitations and tabular analysis. Purchasing parameters and interrelationships. Time source quantity, price, quality, grading systems. Special purchasing systems. Obsolescence .Scrap disposal.

References:-

1. Construction Project Management by KumarNeerajJha
2. Construction Planning and Management by Dr.UKShrivastava
3. Construction Materials, Methods and Techniques by WilliamPSpence.
4. Construction Methods and Measurement by S.W.Nunnally
5. Construction Planning Equipment and Methods by R.Peurifoy



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First YEAR (Semester – 2)

Course Content & Grade

Branch	Subject Title	Subject Code	Contact Hours per Week	Total Credits
M.TECH (Construction Technology & Management)	Financial Management in Construction Industries	MTCTM-2004	3L-1T-0P	4

Course Objective:-

This course will enable students to

CO1:-Understand various aspect of personal management in construction industries.

CO2:- Understand various aspect of Labour management in construction industries.

CO3:- Understand various aspect of waste management in construction industries

CO4:- Understand various aspect of financial management in construction industries

CO5:-Understand the concept of Capital Generation & Financial Accounting Banking

Unit-1 Personnel Management

Principle of personnel management. Qualities of a personnel manager. Objective of personnel management. Personnel policies and programmes. Organizational structure of personnel department. Man power resources. Human resource planning. Job analysis. Performance standards, work rules. Recruitment and selection process. Tests and interview. Induction orientation and indoctrination. Policies, promotion, demotion, transfers etc. Training of personnel. Need for training. Principles of training programmes. Types of training programmes on the job training policy and implementation. Task analysis identification and methodologies.

Unit-2 Labour Management

Industrial relations in construction industry. Principles of industrial relationships. Functional requirements and programme, Industrial disputes, causes of disputes. Types Of disputes. Procedures of the settlement of industrial disputes. Implementation mechanism. Trade Unions - Principles of industrial trade unionism. Objectives and functions. Essentials of trade union.

Unit-3 Waste Management

Introduction to waste and waste management, the concept of productivity and its inter relationship with productivity. Systems concept of waste. Complementarity of waste and resource management. Identification of construction waste material waste, man power waste, energy waste, space waste time waste, equipment waste, capital waste, utilities and services waste.



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Unit-4 Financial Management

Managerial Economics & Financial Statement Nature and scope of managerial economics. Economic theories. Demand analysis and forecasting. Elasticities of demand. Cost and production analysis. Pricing decisions, Policies and practices. Break even analysis. Time value of money, Economics. Comparisons using time value of money basic of comparisons. Decision making amongst alternatives. Cash flow, discounted cash flow. Cash flow forecasting, Project appraisal through financial statements.

Unit-5 Capital Generation & Financial Accounting Banking

Financial Institutes like IFCI, IBI, International financing etc. Book keeping process in construction. The accounting cycle. Journals, ledgers etc. for labour cost, materials and purchases miscellaneous ledgers.

References :-

1. Construction Accounting & Financial Management by Steven J. Peterson
2. Financial Management and Accounting for the Construction Industry Vol. 1 by Matthew Bender
3. Financial Management and Accounting for the Construction Industry Vol. 2 by Matthew Bender
4. Construction Project Management by Paul Netscher
5. Financial Management in Construction Contracting by Andrew Ross & Peter Williams



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Course Content & Grade

Branch	Subject Title	Subject Code	Contact Hours per Week	Total Credits
M.TECH (Construction Technology & Management)	Appropriate Technology and Energy Conservation	MTCTM- 2005(A)	3L-1T-0P	4

Course Objective:-

This course will enable students to

CO1:-Understand various aspect of Appropriate Technology.

CO2:- Understand various aspects of Rural Housing & Rural Environmental Technologies in construction

CO3:- Understand various aspect of Rural Roads in construction.

CO4:- Understand various aspect of Energy Conservation.

CO5:-Understand the concept of use low energy material in construction.

Unit-1 Appropriate technology

Concept and its role in the present circumstances.

Unit-2 Rural Housing & Rural Environmental Technologies

Planning, use of locally available materials, construction techniques. Concept and scope in rural areas planning of water supply schemes in rural areas, development of preferred sources of water, springs, wells, infiltration wells infiltration galleries, collection of rain water, specific problems and method's in rural water supply and treatment. Treatment and disposal of waste water, community and sanitary latrines. Compact and simple waste water disposal systems, biogas plants.

Unit-3 Rural roads

Planning of rural roads, Socio-economic aspects, materials for rural roads, design aspects, drainage problems, and maintenance of rural roads.

Unit-4Energy Conservation

Energy production, distribution and utilization, a review of global situation. Energy Trends, renewable and non renewable sources, research reviews. Building designs and energy factors affecting energy budget in buildings and settlements, Design of buildings for minimising energy. Solar, Wind and Tidal energies, a review and their adoptability.

Unit-5 Low energy materials, construction techniques and environmental control.



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References-

1. Introduction to Energy Technologies for Efficient Power-Generation Alexander V. Dimitrov
2. Energy Efficiency by Ming Yang & Xin Yu
3. Sustainable Energy technologies by Eduardo Rincon Mejia and Alejandro de las Heras
4. Appropriate Technology for Rural Deveiopment by Derek Miles
5. Towards Global Actionb For Appropriate Technology by A.S. Bhalla



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Course Content & Grade

Branch	Subject Title	Subject Code	Contact Hours per Week	Total Credits
M.TECH (Construction Technology & Management)	Sustainable Construction Technology	MTCTM- 2005(B)	3L-1T-0P	4

Course Objective:-

This course will enable students to

CO1:-Understand various aspect of Sustainable Construction Technology.

CO2:- Understand various aspects of Embodied Energy of Materials and Calculations

CO3:- Understand various aspect of Sustainability Implications in Construction Materials

CO4:- Understand various aspect of Sustainability Issues in Metals.

CO5:-Understand the concept of use .Sustainability in Alternate Construction Materials

Unit 1.Introduction to Sustainability: Definition of Sustainability, Resources and Reserves in Construction Materials, Greenhouse Effect and Global Warming, Carbon Footprint of Materials and Calculations.

Unit 2.Embodied Energy of Materials and Calculations: Life Cycle Assessment and Cost Analysis, Sustainability Rating Systems.

Unit 3.Sustainability Implications in Construction Materials: Sustainability Issues in Cement and Concrete Materials, Sustainability Issues in Asphalt Materials.

Unit 4.Sustainability Issues in Metals: Sustainability Issues in Wood, Sustainability Issues in Glass, Sustainability Issues in Masonry.

Unit 5.Sustainability in Alternate Construction Materials: C& D Waste Management Practices, Case Studies and Student Presentations.

COURSE OUTCOMES:

After completion of this course, the student will be able to:

1. Understand the limitations of construction techniques and energy efficient construction.
2. Analyze productivity and economics in construction techniques.
3. Understand reliable proportioning concepts in construction techniques.
4. Implement modular construction practices and sustainable material.



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Reference Books:

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1. Design and Control of Concrete Mixtures, Kosmatka, S.R. and Wilson, M.L., 16th Edition, Portland Cement Association, 2015
2. Hand Book of Green Chemistry and Technology; by James Clarke and Duncan Macquarrie; Blakwell Publishing.
3. Anastas, P. T., Warner, J. C. Green Chemistry: Theory and Practice, Oxford University Press Inc., New York, 1998.
4. Matlack, A. S. Introduction to Green Chemistry Marcel Dekker: New York, NY, 2001



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First YEAR (Semester – 2)

Course Content & Grade

Branch	Subject Title	Subject Code	Contact Hours per Week	Total Credits
M.TECH (Construction Technology & Management)	Construction Costing and Financial Management	MTCTM-2005(C)	3L-1T-0P	4

Course Objective:-

This course will enable students to

CO1:-Understand various aspect of Construction Costing of project.

CO2:- Understand various aspects of cash flow in construction project.

CO3:- Understand various aspects of Cash and Payment of Works in projects

CO4:- Understand the various aspect of material management.

CO5:-. Understand the various aspect of financial management in project.

Unit 1.Construction Costing:

Costing of construction Works; different methods of costing, cost elements in a project; analysis of rates; non-scheduled items of work; cost estimation for a small construction job; purpose, methods and stages of cost control; cost monitoring; cost forecasting methods; variations in individual items of work and their effect on total contract price; valuation of variations. Methods of measurement of earthwork, RCC, Brickwork, Woodwork joinery, steel and iron work plastering/ painting and white/colour washing & painting

Unit 2.Cash Flow:

Determining the funds required for a construction job; preparing cash flow statements; cash inflow and outflow during contract period; project expectations.

Unit 3.Cash and Payment of Works:

Precautions in custody of cash, imprest account and temporary advance; maintenance of temporary advance; and advance account; different types of payment, first, running, advance and final payments.

Unit 4.Material Management:

Objectives and scope of material management classification, codification, ABC analysis, standardization and substitution; introduction to inventory control; stores management; organization and lay out; receipt, inspection and issue; care and safety; store records and store accounting.



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Unit 5.Financial Management

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Meaning and scope; financial statement analysis; funds flow analysis; capital budgeting; cost-benefit analysis.

COURSE OUTCOMES:

After completion of this course, the student will be able to:

1. Learn basics of construction costing, cash flow analysis along with their practical applications.
2. Acquire good knowledge of the management of various materials at the site.
3. Learn capital budgeting, cost benefit analysis, financial management along with their field applications

Reference Books:

1. Mueller, F.W. Integrated cost and schedule control for construction projects.
2. Gobourne: Cost control in the construction industry.
3. Schedule of rates, specification manuals etc. from PWD
4. Chris Hendrickson and Tung Au: Project Management for Construction
5. Datta: Material Management Procedures, Text and Cases, 2e Prentice Hall
6. Gopalakrishnan, P, Sundaresan, M: Material Management - an Integrated Approach, Prentice Hall.
7. Dobbler and Bart: Purchasing and Supplies Management, Text and Cases, 6e
8. Chitkara, K.K.: Construction Project Management; Tata-McGraw Hil.



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Course Content & Grade

Branch	Subject Title	Subject Code	Contact Hours per Week	Total Credits
M.TECH (Construction Technology & Management)	CONSTRUCTION DEMOLITION AND WASTE MANAGEMENT	MTCTM- 2005(D)	3L-1T-0P	4

Course objectives:

This course will enable students

- focus on the principles of sustainable construction and demolition waste management and resource efficiency
- examining the environmental impact of building materials; formulating and designing pre-construction and site waste management plans

Syllabus Contents:

Unit 1:

Environmental Impact of Building Materials Embodied energy of materials; impact on the local environment; toxicity of the material; life cycle assessment. Nature and Source Direct and indirect waste; site types and origins; composition; quantity; current recycling/reuse potential of building materials..

Unit 2:

Construction and Demolition Waste Management Plans International good practice; planning requirements; DoEHLG guidance document; company policy; demolition plans; site implementation; supplier agreements; sub-contractor management; role of waste management contractor; training; auditing; skip management; current markets; current disposal options; health and safety; reporting to local authorities. Treatment of Construction and Demolition Waste, waste permits; waste licenses; waste transfer facilities; landfills; treatment technologies; hazardous waste facilities; reporting to EPA.

Unit 3:

Designing for Waste Prevention and Minimisation Waste prevention and minimization; client, contractor and designer attitudes; proper maintenance of existing buildings; reuse of existing building structure; design flexibility; design for reuse and recycling; dimensional co-ordination and standardization; modular design; material selection and control.

Unit 4:



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Waste Forecasting Tools Application of WRAP's designing out waste tool for buildings and civil engineering; WRAP net waste tool; BRE SMARTWaste; WRAP Site Waste Management Plan Tracker

Unit 5:

Future developments Potential future markets; 'smart' materials; use of eco-materials.

Course outcomes:

On completion of this course, students are able to:

- Formulate, design, evaluate and review pre-construction and construction phase resource efficient waste management plans
- Evaluate, assess and recommend potential reuse/recycling/disposal options considering existing and potential future markets/uses.

REFERENCES:

1. Springer, "Recycling and Resource Recovery Engineering", Springer-Verlag Berlin Heidelberg (1996)
2. Greg Winkler, "Recycling Construction and Demolition waste: A LEED-Based Toolkit (Green Source) (Google ebook), Mc Graw Hill Professional
3. V M Tam, Chi Ming Tam, "Reuse of Construction and Demolition Waste in Housing Development", Nova Science Publishers, 2008



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Course Code: 6TMST-204

Course Content & Grade

Branch	Subject Title	Subject Code	Contact Hours per Week	Total Credits
M.TECH (Construction Technology & Management)	LAB-III :-NDT-LAB	MTCTM-2006	3L-1T-0P	4

Course Code: 6TMST-205

Course Outcomes

After studying this subject, the students will be able to:

CO1: Perform the various non-destructive tests on concrete.

List of Experiments:

Laboratory work shall be based on the topics of Non-destructive testing and as per the facility available in the institution.

1. To find compressive strength of concrete using Schmidt Rebound Hammer.
2. To calibrate Rebound Hammer.
3. To check the quality of concrete in the structures using Ultrasonic Pulse Velocity meter.
4. To measure the depth of cover in RCC structures using cover meter.
5. To detect corrosion in reinforcement by Half Cell Potentiometer.
6. To determine the permeability of cement mortar and concrete in RCC structures.
7. To find rate of corrosion in reinforcement in RCC structures.
8. To find compressive strength of concrete using CAPO test.
9. To perform various destructive tests on concrete specimen.
10. To perform chloride test on RCC structures.
11. To perform carbonation test on RCC structures.



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Course Content & Grade

Branch	Subject Title	Subject Code	Contact Hours per Week	Total Credits
M.TECH (Construction Technology & Management)	Lab-IV Cont. Mgmt. Software Lab	MTCTM-2007	3L-1T-0P	4

Course objectives:

This course will enable students to

- The objective of this course is to make students to learn principles and design of structures.

Courses content

Software Application Use of construction management softwares (MS-PROJECTS, PRIMAVERA), preparing of estimation of a structure using excel Analysis of skeletal and continuum structures using standard FEM packages, BIM

Course outcomes:

On completion of this course, students are able to:

- **Achieve Knowledge of Design and development of soft skills.**
- **Understand the principles of design of structures using the standard packages.**

REFERENCES:

1. “Software Manuals”
2. Harry G Harris and Gajanan M Sabnis, “Structural Modeling and Experimental Techniques”, CRC Press